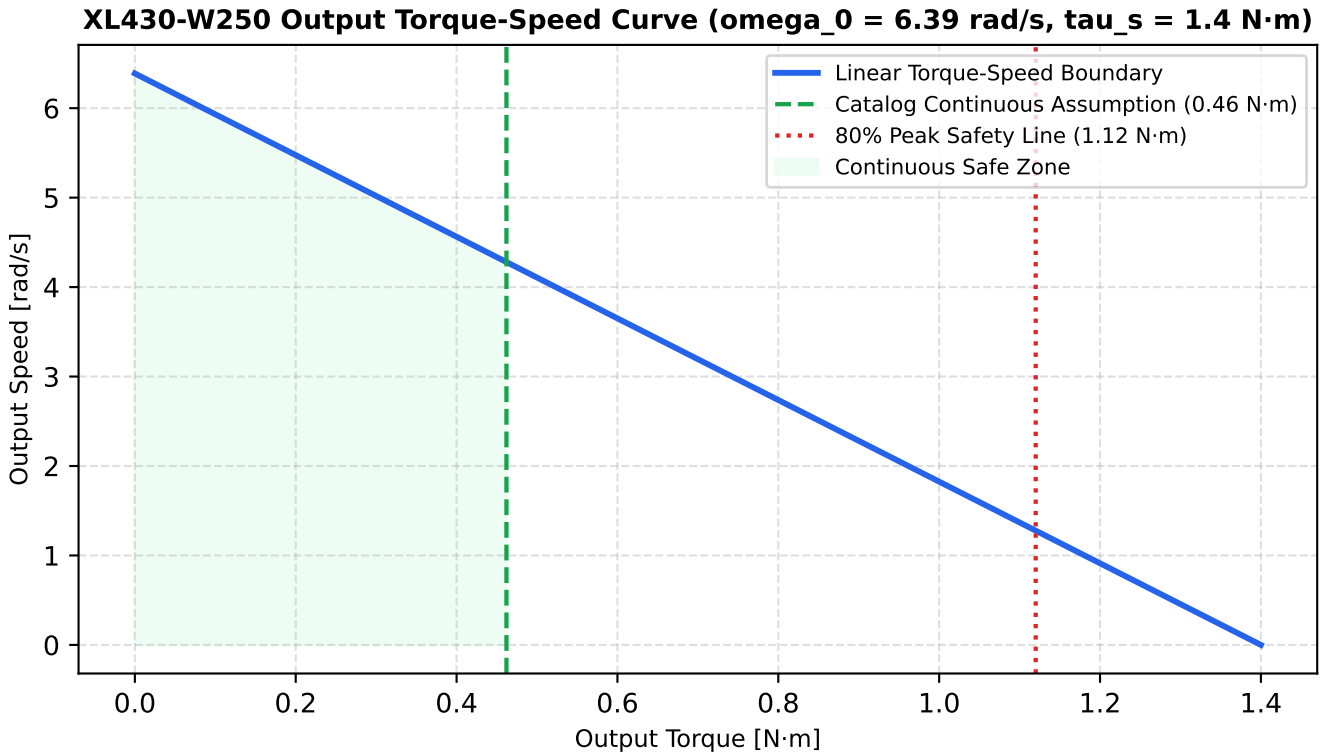
Part Number: XL430-W250

1. Electrical & Mechanical Specifications (Standard Reference)

Parameter	Symbol	Specification Value	Test Condition / Notes
Part Number	ID	XL430-W250	DYNAMIXEL-X
Manufacturer	Vendor	ROBOTIS	Cored DC Motor
Rated Input Voltage	V_rated	12.0 V	Nominal DC Bus
Stall / Peak Torque	tau_stall	1.4 N.m (at 12.0V, 1.3A)	At 12.0 V
Continuous Rated Torque	tau_cont	Not specified (use thermal der	25 degC ambient, continuous duty
No-Load Speed	omega_0	61 rev/min (at 12.0V)	Unloaded output shaft
Actuator Mass (Weight)	m_act	57.2 g	Includes housing & internal gearhead
Internal Reduction	i_int	258.5 : 1	Integrated gear stage
Envelope Dimensions	W x H x D	28.5 x 46.5 x 34.0 mm	Outer housing envelope



2. Mechanical Mounting & Frame Integration
   - Housing Envelope: 28.5 mm x 46.5 mm x 34.0 mm
   - Catalog Mass (unverified): 57.2 g (0.0572 kg)
   - Mounting Pattern: Not verified; obtain manufacturer flange/shaft drawings.

3. Thermal & Duty-Cycle Notes
   - Catalog continuous torque is an input assumption until its manufacturer rating is check
   - The 0.33x stall estimate, where used, is a heuristic rather than a measured thermal rat
   - The linear torque-speed envelope is a screening model, not a measured performance curve

4. Project Reference Index
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